

Hello everyone! Welcome to the lab! Today, we are diving into the geometry of power. We're talking about **gerrymandering**, but not just the "look at this crazy-shaped district" kind. We are talking about the **math of the Efficiency Gap**.

This isn't just an abstract number—it's a "single, tidy number" that captures the packing and cracking decisions that go into a district plan. Let's get to work!

Part 1: The Core Concept—Wasted Votes (15 Minutes)

To understand the Efficiency Gap, you have to understand the **"Wasted Vote."** In our winner-take-all system, a vote is mathematically "inefficient" if it doesn't directly contribute to a victory.

There are two ways to waste a vote:

1. **Lost Votes:** Any vote cast for a losing candidate. (This is the result of **"cracking"** a group of voters so they can't reach a majority).
2. **Surplus Votes:** Any vote for a winning candidate *in excess* of the 50% plus one needed to win. (This is the result of **"packing"** voters into a single district to dilute their power elsewhere).

The Definition: The Efficiency Gap (EG) is simply the difference between the parties' respective wasted votes, divided by the total number of votes cast.

Mini-Activity 1: The "10-District" Challenge (10 Minutes)

The Scenario: You have a state with 10 districts. Each has 100 voters.

- **Party A** wins 55% of the statewide vote (550 votes total).
- **Party A** wins 70 votes in Districts 1–3.
- **Party A** wins 54 votes in Districts 4–8.
- **Party A** wins 35 votes in Districts 9–10.

Your Task: Calculate the wasted votes for both parties. (*Professor's Tip: Assume you only need 50 votes to win for simplicity.*)

Solution:

- **Party A** wasted votes: $20 \text{ (D1-3)} + 4 \text{ (D4-8)} + 35 \text{ (D9-10)} = \mathbf{150}$.
- **Party B** wasted votes: $30 \text{ (D1-3)} + 46 \text{ (D4-8)} + 15 \text{ (D9-10)} = \mathbf{350}$.
- **The Gap:** $(350 - 150) / 1,000 = \mathbf{20\%}$.

Part 2: The Math Professor's Shortcut (15 Minutes)

Summing every district is tedious. If districts are equal in population and it's a two-party race, we use this beautiful formula:

Efficiency Gap = Seat Margin – (2 × Vote Margin).

- **Seat Margin:** The share of seats won minus 50%.
- **Vote Margin:** The share of votes won minus 50%.

In our activity above:

- Party A won 8 of 10 seats (80%). Seat Margin = 30% (0.30).
- Party A won 55% of the vote. Vote Margin = 5% (0.05).
- **EG = 0.30 - (2 × 0.05) = 20%.**

The 2-to-1 Rule: For a plan to be "fair" (EG of zero), every 1% increase in vote share should result in a **2% increase in seat share**. This accounts for the "winner's bonus" common in American politics.

Part 3: Efficiency Gap vs. Partisan Bias (15 Minutes)

Before the EG, scholars used "**Partisan Bias**." But Partisan Bias has a major flaw: it requires a **hypothetical election**. You have to "shift" the votes to imagine what would happen in a 50-50 tie.

The Efficiency Gap is superior for three reasons:

1. **Actual Results:** It uses real data, not counterfactuals.
2. **No "Uniform Swing":** Partisan bias assumes every district shifts by the same percentage, which is often geographically inaccurate.
3. **Uncompetitive States:** You can calculate the EG even in lopsided states (like Utah or Massachusetts) where a 50-50 tie is "fanciful" and impossible to model.

Mini-Activity 2: The "When to Use It" Debate (10 Minutes)

I'm giving you a list of scenarios. Based on the sources, tell me if the Efficiency Gap is the right tool or if we need to be careful.

1. **A state where Party A wins 80% of the vote.**

- *Answer:* Be careful! If a party wins more than 75%, additional votes look "wasted" even if they already own every seat.
 - 2. **A plan that looks fair today but might change.**
 - *Answer:* Use **Sensitivity Testing**. Shift the vote share by 5-7% to see if the gap "crosses the zero axis".
 - 3. **A race where one candidate ran unopposed.**
 - *Answer:* Use **Imputation**. Don't use 100% of the vote; assign a value (like 70/30) based on how that district usually votes in presidential races.
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Part 4: Doctrine and Limits (10 Minutes)

So, how much is "too much"? The creators of the EG propose specific **thresholds**:

- **Congressional Plans:** A gap of **2 seats**.
- **State House Plans:** A gap of **8 percent**.

The Final Caveat: A high EG doesn't always mean "illegal." A state can defend itself by showing the gap was **inevitable** because of political geography (e.g., Democrats clustering in cities) or because they were following other legitimate policies like keeping counties together.

Conclusion: The Efficiency Gap isn't just a formula; it's a way to "slice the Gordian knot" of gerrymandering. Class dismissed—go crunch some numbers!

Alright class, welcome back from that activity! Now that you've mastered the basic calculation, we need to put on our "skeptical mathematician" hats. While the **Efficiency Gap (EG)** is a "single, tidy number" that captures packing and cracking, it is far from a perfect crystal ball.

In fact, some mathematicians argue that if we aren't careful, this formula could take on a "life of its own" in courtrooms. Let's dive into the **mathematical critiques** and the **safety warnings** for using this tool.

Part 5: The Mathematician's Critique (15 Minutes)

First, let's look at what happens when we simplify the math. Under the assumption of equal turnout across districts, the EG reduces to a very simple relationship: $EG = \text{Vote Lean} - 0.5 \times \text{Seat Lean}$ (or $\tau - 0.5\sigma$). This reveal is a bit of a bombshell because it shows that the EG is actually just a **double-proportionality standard**.

Here are the four big "Red Flags" that mathematicians Bernstein and Duchin point out:

1. **The Proportionality Paradox:** Most people think "fairness" means if you get 60% of the votes, you should get 60% of the seats. **The EG disagrees.** If a party wins 60% of the vote and 60% of the seats, the EG actually flags this as a gerrymander favoring the *other* party! It mandates that for a plan to be "fair" (EG = 0), the seat lean must be **exactly twice** the vote lean.
 2. **Fetishizing Landslides:** The EG is "perfectly neutral" only when a district has a 75-25 split. Why? Because in a 75-25 race, both sides waste exactly 25% of the total vote. This means the formula actually **rewards non-competitive landslide districts** and penalizes competitive ones.
 3. **The "Edge Case" Breakdown:** If a party is *too* popular—specifically over 75% of the statewide vote—the math breaks. At 80% vote share, it is mathematically impossible to have a "fair" EG; the formula will label *any* map as a gerrymander against the minority party.
 4. **The Granularity Problem:** In small states with only a few districts, the EG is "too coarse". If a state only has two or three seats, there are only a handful of possible EG values, making it nearly impossible to hit that "fair" threshold.
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Part 6: When Should We (and Shouldn't We) Use It? (10 Minutes)

Because of these quirks, the sources agree that the EG should never be a "dispositive" or standalone indicator. It's a **corroborative tool**, not the whole trial. Here are the **caveats** for its appropriate use:

- **The "Rule of Eight":** Analysts generally only use the EG for states with **eight or more districts** to ensure the math is "granular" enough to be meaningful.
 - **Sensitivity Testing is Mandatory:** You cannot just look at one election result. You must perform a "sensitivity analysis"—shifting the vote share by roughly 5% to 7%—to see if the partisan advantage is **durable** or just a result of a one-time "uniform swing".
 - **The Intent-Effect-Justification Tripod:** In the Wisconsin case, the court ruled that an EG score is only part of a three-pronged test. You need to prove **discriminatory intent** (they *meant* to rig it), **discriminatory effect** (the EG shows they *did* rig it), and a lack of **legitimate justification** (there wasn't a non-partisan reason, like geography, for the map's shape).
 - **Watch for "Uncontested" Noise:** If a race is uncontested, the EG requires "imputation"—essentially making an educated guess about how people would have voted—which adds a layer of speculation to the math.
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Final Activity: The "Expert Witness" Simulation (15 Minutes)

The Scenario: You are testifying in court. A state has 5 districts. The majority party won 60% of the vote and 100% of the seats. The EG is high.

Your Task: Based on our lecture, prepare a 2-minute "cross-examination" response.

- **Question:** "Professor, why should we trust this 8% threshold in a 5-district state?"
 - **Hint:** Think about **granularity** and **small sample sizes**.
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Professor's Closing Thought: Class, the Efficiency Gap is a powerful way to "slice the Gordian knot" of gerrymandering, but it is a tool, not a total solution. As computing power grows, we are moving toward **sampling tools**—where we compare a map against thousands of computer-generated alternatives to see if the outcome is truly an outlier.

But for now, the EG remains the "tidy number" that changed the legal game. See you next week when we talk about the **Geometry of Compactness**! Class dismissed!.

Alright, class, final stretch! We've got the math and we've got the skepticism, but now we need to look at how this "single, tidy number" actually holds up in a court of law. Because, let's be honest, a formula is just a bunch of symbols until a judge says it's a **judicially manageable standard**.

Grab your legal pads. Here is the **History of the Partisan Gerrymandering Cause of Action** and where the Efficiency Gap (EG) fits into the legal landscape.

Part 7: The "Relentless Pessimism" Era (15 Minutes)

Before the Efficiency Gap, the legal story of gerrymandering was, in the words of legal scholars, "relentlessly pessimistic".

1. **Recognition (1983-1986):** The Supreme Court didn't even recognize gerrymandering as something they could rule on until the 1980s. In ***Karcher v. Daggett* (1983)**, Justice Stevens first wrote explicitly about it in a concurrence, suggesting we look for "objective indicia of irregularity".
2. **The Bandemer Trap (1986):** In ***Davis v. Bandemer***, the Court finally said, "Yes, gerrymandering is **justiciable** (we can hear the case)". But they set the bar impossibly high. Plaintiffs had to prove that a plan would "consistently degrade" a group of voters' influence over the process *as a whole*.
 - **The Result:** Over the next 20 years, the record for plaintiffs was **zero wins and fifty losses**. Courts essentially said, "We'll know it when we see it," but they never seemed to see it.
3. **The Scalia/Kennedy Split (2004):** By the time we hit ***Vieth v. Jubelirer***, the conservative plurality (led by Scalia) wanted to give up entirely and declare it a "nonjusticiable political question". But **Justice Kennedy**—our perennial swing voter—refused. He said he wouldn't close the door if someone could find "principled, well-accepted rules of fairness" to measure the burden on representational rights.

Part 8: The Doctrinal Opportunity—*LULAC* (10 Minutes)

In 2006, the case *LULAC v. Perry* created a "sea change". Several justices expressed surprise enthusiasm for **partisan symmetry**—the idea that a plan should treat parties equally in terms of converting votes to seats.

However, Justice Kennedy had **three big hang-ups** with the symmetry measures (like Partisan Bias) used at the time:

1. **Conjecture:** It depended on guessing where "vote-switchers" lived.
2. **Hypotheticals:** He was wary of invalidating a map based on a "hypothetical state of affairs" (like a simulated 50-50 tie).
3. **No Threshold:** He didn't know how much partisan dominance was "too much".

The Efficiency Gap was designed specifically to answer Kennedy's call. It uses **actual election results** (no conjecture), avoids hypothetical simulations, and provides a **clear numeric threshold** (2 seats for Congress, 8% for state houses).

Mini-Activity 3: The "Justification" Rebuttal (10 Minutes)

The Scenario: You are the State's lawyer. A map has an Efficiency Gap of 12% (well over the 8% threshold). **Your Task:** Based on the **Stephanopoulos-McGhee test**, what are the two ways you can defend this map so it isn't struck down?

- **Answer 1:** Prove the gap is the necessary result of a **legitimate, consistently applied state policy** (like keeping counties together or complying with the Voting Rights Act).
 - **Answer 2:** Show the gap was **inevitable** due to the state's **political geography** (e.g., your voters are just naturally "packed" into cities).
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Part 9: Legal Analysis—When is it Useful? (10 Minutes)

So, when does the Efficiency Gap actually "work" in court?

- **The "Rule of Eight":** It is most useful in states with **eight or more districts**. In small states like Idaho or Rhode Island, the math is too "coarse" and "nongranular"—a single seat flip can swing the EG wildly, making it an unreliable legal standard.
- **The "Intent-Effect-Justification" Tripod:** In the **Wisconsin case (*Whitford v. Gill*)**, the court used EG as part of a three-pronged test: 1) **Discriminatory Intent** (did they mean

to rig it?), 2) **Discriminatory Effect** (does the EG show a large, durable gap?), and 3) **Lack of Justification** (could they have drawn a fairer map?).

- **As "Corroborative Evidence":** The most recent legal analysis suggests EG should be an **indicative measure**, not the *exclusive* definition of gerrymandering. It's a "starting point" to show when partisan advantage has risen to the level of **vote dilution**.

The Final Caveat: Legal Blind Spots

The Efficiency Gap has a major legal "perversity": it is **blind to voter suppression**. If a party suppresses the other side's turnout, the EG might actually *decrease*, making a gerrymander look "fairer" because the suppressed party now has fewer wasted votes.

Conclusion: Class, the Efficiency Gap is the math community's attempt to "slice the Gordian knot" the Supreme Court tied for itself. It's a powerful tool for catching the "most extreme gerrymanders in modern history", but as the law evolves, it will likely be just one tool in a larger kit that includes computer-generated sampling and algorithmic comparisons.

Class dismissed! Go out there and make the math count!